

# Arquitectura Medallón

Ingesta, calidad e indexación semántica del catálogo de AniList

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[github.com/jesúsarredondo98/Pipeline\\_Medallion\\_Architecture\\_AnimeIndex](https://github.com/jesúsarredondo98/Pipeline_Medallion_Architecture_AnimeIndex)  
Requerimientos: PRD v2.4 · Harness: Agents.md

25

SKILLS ATÓMICAS

19/19

PRUEBAS PASANDO

994/1000

REGISTROS VÁLIDOS EN SILVER

1 Arquitectura

AniList API

Bronze

Silver

Gold

Búsqueda

POST paginado · rate-limit gobernado (pausa base 1s + frenado proactivo) · reintentos con backoff

/bronze/\*.json  
crudo + metadata de ingesta únicamente · checkpoint por archivo existente

/silver/anime (Delta Lake) · contrato Pydantic V2 · limpieza determinista · MERGE idempotente por id

/gold/index · embeddings sólo de description · benchmark FAISS/usearch · sólo se embebe el delta

Texto libre → embedding → vecinos más cercanos → id / título / score / sinopsis

2 Decisiones clave de Fase 0 / Fase 1

- **Python 3.12** sobre imagen `linux/arm64` — cobertura de wheels precompiladas confirmada para las 8 dependencias auditadas; ninguna se compila desde fuente.
- `hnswlib` **sustituido por** `usearch` : se verificó que hnswlib nunca publicó wheels para ninguna plataforma (sólo sdist) en sus 8 versiones históricas.
- `torch` **CPU** instalado desde el índice dedicado ( `download.pytorch.org/whl/cpu` ) en vez del paquete estándar de PyPI, que arrastra el stack CUDA completo incluso en arm64 sin GPU.
- **Motor vectorial de producción: FAISS** — decidido por benchmark real contra el corpus, no supuesto. Regla de decisión: recall exacto pesa más que latencia a este volumen; en empate, gana la menor huella en disco.

Guard de fuga de título retirado en description-embedder : una verificación de contenido en tiempo de ejecución produjo un falso positivo real (la sinopsis de *Eyeshield 21* menciona ese nombre de forma legítima). La invariante se garantiza estructuralmente y se verifica por inspección de código.

### 3 Catálogo de skills (25)

#### TRANSVERSAL

- repo-hygiene-scaffold
- docker-arm64-scaffold
- arm64-dependency-audit

#### BRONZE — INGESTA

- anilist-graphql-client
- rate-limit-governor
- http-retry-policy
- bronze-batch-writer
- bronze-resume-checkpoint
- failed-pages-ledger

#### SILVER — CALIDAD

- html-description-normalizer
- anime-data-contract
- quarantine-writer
- content-hash-builder
- delta-merge-upsert
- silver-metrics-emitter

#### GOLD — IA

- vector-engine-benchmark
- description-embedder
- vector-index-builder
- gold-delta-indexer
- semantic-search-api

#### PRUEBAS Y REPORTE

- contract-test-kit
- idempotency-test-kit
- resilience-test-kit
- semantic-test-kit
- execution-report-renderer

### 4 Resultados de ejecución

#### 4.1 Idempotencia de Silver (REQ-S4) — 20 páginas reales de AniList

| Métrica              | Corrida 1 | Corrida 2 |
|----------------------|-----------|-----------|
| filas_leidas         | 1000      | 1000      |
| filas_validas        | 994       | 994       |
| filas_en_cuarentena  | 6         | 6         |
| filas_nuevas         | 994       | 0         |
| filas_actualizadas   | 0         | 0         |
| filas_totales_silver | 994       | 994       |

La segunda corrida sobre el mismo lote produce cero inserciones y cero actualizaciones — el MERGE nativo de Delta Lake confirma la idempotencia sin recalcular nada a mano.

#### 4.2 Comparativa de motores vectoriales (REQ-G0)

| Motor                                  | Construcción (ms) | Latencia p50 (ms) | Latencia p95 (ms) | Recall@k | Huella en disco |
|----------------------------------------|-------------------|-------------------|-------------------|----------|-----------------|
| faiss.IndexFlatIP <span>ELEGIDO</span> | 0.42              | 0.069             | 0.087             | 1.000    | 1,526,829 B     |
| usearch.Index (HNSW)                   | 30.09             | 0.079             | 0.089             | 1.000    | 1,673,840 B     |

Empate en recall@k (1.000) contra fuerza bruta real — gana FAISS por menor huella en disco, siguiendo la regla de decisión del PRD.

### 4.3 Suite de pruebas — 19/19

| Categoría             | Cubre                                                                          | Resultado |
|-----------------------|--------------------------------------------------------------------------------|-----------|
| Contrato              | id ausente / description vacía → cuarentena con motivo correcto                | 4/4       |
| Idempotencia (Silver) | doble corrida sin duplicados, content_hash estable, updates reales             | 3/3       |
| Resiliencia (Bronze)  | 429/Retry-After, ritmo proactivo, reanudación sin re-descarga, fallo no aborta | 5/5       |
| Semántica + Gold      | búsqueda válida, idempotencia en Gold, delta-only, benchmark real              | 7/7       |

Verificado en dos entornos: venv local (Python 3.11, macOS arm64) y contenedor Docker real (Python 3.12, linux/arm64).

Cada consulta se embebe con `all-MiniLM-L6-v2` a partir únicamente del texto del prompt, y se compara contra los 994 vectores de descripción de Silver (nunca contra el título). La columna *synopsis* muestra el texto real del anime encontrado.

| “pirates searching for treasure” |                                                   |        |                                                                                                                                                                     |
|----------------------------------|---------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| id                               | title_romaji                                      | score  | synopsis                                                                                                                                                            |
| 465                              | ONE PIECE THE MOVIE: Karakuri-jou no Mecha Kyohei | 0.6065 | The crew salvages a treasure chest from a sinking wreck, but inside turns out to be an old lady hiding. To get the Straw Hat Pirates to take her home, she promi... |
| 459                              | ONE PIECE (Movie)                                 | 0.5959 | There once was a pirate known as the Great Gold Pirate Woonan, who obtained almost 1/3 of the world's gold. Over the course of a few years, the pirate's existen... |
| 21                               | ONE PIECE                                         | 0.5337 | Gold Roger was known as the Pirate King, the strongest and most infamous being to have sailed the Grand Line. The capture and death of Roger by the World Govern... |
| 462                              | ONE PIECE THE MOVIE: Dead End no Bouken           | 0.4663 | Luffy and crew arrive at the harbour of Anabaru. The local casino is holding a competition in which the winner will obtain a huge monetary reward if he reaches...  |
| 602                              | KO Seiki Beast Sanjuushi                          | 0.4509 | Who cares about saving the world when there's treasure to be found! When Wan and his voracious appetite hook up with bossy little Mei-Mer and playboy Bud Mint,...  |

| “mecha pilots defending a city” |                                 |        |                                                                                                                                                                     |
|---------------------------------|---------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| id                              | title_romaji                    | score  | synopsis                                                                                                                                                            |
| 218                             | Kidou Senkan Nadesico           | 0.4849 | Akito doesn't want to fight. Despite a childhood spent on the anime Gekiganger 3, a Mecha show, he'd rather cook than pilot a Mecha. Fate intervenes when his ho... |
| 764                             | Zoids Shin Seiki/Zero           | 0.4235 | Zoids - powerful animal-shaped combat mechs - are no longer used in warfare, but in organized sporting competitions. The Blitz Team, a group of pilots strugglin... |
| 608                             | Sakura Taisen: Katsudou Shashin | 0.4204 | Like her father before her, Sakura was born to defend her people against the Black Sanctum Council. But her dreams of being a hero take a strange turn when she...  |
| 769                             | BURN-UP W                       | 0.4179 | When crime and terrorism go out of control in Tokyo, the police unleash Team Warrior - a band of highly-skilled and completely reckless band of female cops. Fea... |
| 878                             | Zegapain                        | 0.4161 | Kyo, an avid swimmer tries his best to keep the high school swimming club going, which is pretty hard when he's the only member. Facing the prospect of the club... |

### “detective solving a murder mystery”

| id  | title_romaji              | score  | synopsis                                                                                                                                                            |
|-----|---------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 220 | Kuro no Danshou           | 0.5478 | Based on the erotic game by Abogado Powers & Oz Club. While vacationing with his adopted daughter, Private Detective Susuki becomes trapped in a secluded ski re... |
| 235 | Meitantei Conan           | 0.4956 | Kudo Shinichi is a seventeen year-old high school detective whom people call the "Modern Sherlock Holmes." However, one night after a date with his childhood sw... |
| 493 | Armitage III: POLY-MATRIX | 0.4351 | In the year 2046 humans are moving away from using robots and begin to trust them less and less. In this time life becomes very perilous for the beingd known as... |
| 365 | FAKE                      | 0.4349 | Dee Laytner and Randy "Ryo" Maclane are two New York cops taking a needed vacation in England. Dee has very strong feelings for Ryo and wants to take this chanc... |
| 301 | Cinderella Boy            | 0.4301 | Ranma is a poor private detective with a strong sense of justice. His partner, Rella, is a thrill-seeking rich girl. They make for an odd couple on the dangerou... |

### “high school romance and friendship”

| id  | title_romaji             | score  | synopsis                                                                                                                                                            |
|-----|--------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 472 | ToHeart                  | 0.5275 | For as long as Akari can remember, she and Hiroyuki have always been friends. But with time, everything changes, and her feelings have turned into something mor... |
| 730 | Houkago no Shokuinshitsu | 0.5091 | Two teachers and their ups and downs of being in love with each other.                                                                                              |
| 863 | Joshikousei              | 0.4637 | Eriko and her friends Yuma and Ayano are excited about entering high school. Their excitement leads to their breaking of the rules when they toured the school b... |
| 105 | BOYS BE ...              | 0.4611 | There are two sides to every love story, and Boys Be... reveals what's really going on - from the guy's point of view! Meet three normal high school guys with j... |
| 743 | Umi ga Kikoeru           | 0.4590 | Ocean Waves is high school love story set in Kochi, Japan. Morisaki Taku and Matsuno Yutaka are best friends when a girl from Tokyo named Muto Rikako transfers...  |

### “space battle between rival factions”

| id   | title_romaji             | score  | synopsis                                                                                                                                                            |
|------|--------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 820  | Ginga Eiyuu Densetsu     | 0.5824 | For decades, the Galactic Empire has been locked in an interstellar war with the Free Planets Alliance, a conflict that involves thousands of spaceships and mil... |
| 274  | Kiddy Grade              | 0.4292 | After fighting in numerous interstellar wars, mankind forms the Galactic Union to bring stability and to prevent conflicts between different planets. However, d... |
| 1090 | Kidou Senshi Gundam I    | 0.4199 | It is the year Universal Century 0079. Over 90 million humans populate the Earth, and many have been forced to live in space. People now live in large cylinder-... |
| 865  | Kikou Senki Dragonar     | 0.4179 | A.D. 2087 - the United Lunar Empire Giganos wages war on the Earth Federation Military to take control of the planet and establish a "rebirth" of the human race... |
| 1088 | Choujikuu Yousai Macross | 0.4119 | In 1999, a giant alien spacecraft crash lands on South Ataria island. Humanity proceeds to attempt to rebuild this marvel, but political conflicts centered arou... |

**“cooking competition between chefs”**

| id  | title_romaji                                           | score  | synopsis                                                                                                                                                            |
|-----|--------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 110 | Chuuka Ichiban!                                        | 0.4340 | The story takes place in 19th century China during the Qing Dynasty, where the Emperor was weakened and the country was close to chaos. It is also during a fict... |
| 216 | Pia Carrot e Youkoso!!:<br>Sayaka no Koi<br>Monogatari | 0.3158 | High schoolers Sayaka Takai and Akihiko Kannazuki work together at a restaurant chain "Pia Carrot." Both are interested in each other, but before either could m... |
| 645 | Hikaru no Go: Hokuto<br>Hai e no Michi                 | 0.2985 | Hikaru is given a phone call asking him to appear in the Japan/China/Korea Junior's Tournament but he finds out he will have to take place in a preliminary tour... |
| 314 | Ace wo Nerae! 2                                        | 0.2768 | Three years has passed since the events of the first series. Hiromi continued training with her coach and became Japan's junior champion. Their common profits h... |
| 808 | Ganso Bakuretsu<br>Hunters                             | 0.2766 | Take two gorgeous women with whips, add a magician and a body builder, tie it all together with an amorous young man willing to put the moves on anything that m... |

**“a lonely robot learning about humanity”**

| id  | title_romaji                                      | score  | synopsis                                                                                                                                                            |
|-----|---------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 475 | Hotori: Tada Saiwai wo<br>Koinegau                | 0.5054 | At the Personality Plant, robots are being built and slowly outfitted with the artificial memories of real people. Suzu is one such robot, being made to replace... |
| 493 | Armitage III: POLY-<br>MATRIX                     | 0.4583 | In the year 2046 humans are moving away from using robots and begin to trust them less and less. In this time life becomes very perilous for the beingd known as... |
| 976 | Yokohama Kaidashi<br>Kikou: Quiet Country<br>Cafe | 0.4216 | In the near-future Japan, global warming has brought the large city Yokohama underwater, and only the hills remain above the ocean surface. What used to be one...  |
| 949 | Top wo Nerae!<br>GunBuster                        | 0.3986 | In the very near future, a race of huge, insect like aliens is discovered traveling the galaxy. These aliens seem dedicated to the eradication of the human spec... |
| 573 | Saber Marionette J                                | 0.3945 | In the distant future, since the Earth has become overpopulated, efforts to find and colonize on other planets have begun. However, one of the ships, the "Mesop... |

**“a young ninja who dreams of becoming the leader of his village”**

| id  | title_romaji                                                 | score  | synopsis                                                                                                                                                            |
|-----|--------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20  | NARUTO                                                       | 0.5281 | Naruto Uzumaki, a hyperactive and knuckle-headed ninja, lives in Konohagakure, the Hidden Leaf village. Moments prior to his birth, a huge demon known as the Ky... |
| 229 | Ninin ga Shinobuden                                          | 0.5269 | Kaede is a normal school girl who was studying for her exams for school when suddenly she is interrupted by Shinobu, a girl who is a ninja-in-training, attempti... |
| 594 | NARUTO: Takigakure no<br>Shitou - Ore ga Eiyuu<br>Dattebayo! | 0.5122 | A routine rank-C mission turned into a full-blown battle as the Hidden Fall village is attacked by enemy ninjas. Now Naruto, Sasuke and Sakura must help the lea... |
| 48  | .hack//SIGN                                                  | 0.5064 | A young wavemaster, only known by the alias of Tsukasa, wakes up in an MMORPG called The World, with slight amnesia. He does not know what he has previously don... |
| 967 | Hokuto no Ken                                                | 0.5063 | Nuclear war has devastated the planet, turning one fertile ground into a barren, inhospitable landscape on which mankind struggles desperately to stay alive. So... |

“a young boy with superhuman strength searching for magical dragon balls that grant wishes”

| id  | title_romaji                                                | score  | synopsis                                                                                                                                                             |
|-----|-------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 223 | Dragon Ball                                                 | 0.5938 | Goku Son is a young boy who lives in the woods all alone—that is until a girl named Bulma runs into him in her search for a set of magical objects called the "D...  |
| 502 | Dragon Ball: Shen Long no Densetsu                          | 0.4922 | A retelling of Dragon Ball's origins, this is a different version of the meeting of Goku, Bulma, Oolong, and Yamucha. They are all looking for the dragon balls...   |
| 796 | Mahoujin Guru Guru Movie                                    | 0.4915 | The hero Nike and magic user Kukuri learn of the "Pickle of Happiness", an item that will grant a wish to the one who eats it. With the help of old friends, the...  |
| 225 | Dragon Ball GT                                              | 0.4893 | Emperor Pilaf finally has his hands on the Black Star Dragon Balls after years of searching, which are said to be twice as powerful as Earth's normal ones. Pila...  |
| 987 | Dragon Ball GT: Goku Gaiden! Yuuki no Akashi wa Suushinchuu | 0.4642 | Years after the end of the Dragonball GT, the story continues in this special with Son Goku's now elderly granddaughter Pan, and a new generation of super saiyaj... |